SMALL EIGENVALUES ON DEGENERATING CURVES AND THE INTERSECTION FORM

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Abstract. Let a holomorphic family of compact Riemann surfaces over a disc, with smooth total space, degenerate to a reduced curve with N irreducible components and arbitrary singularities, and let the fibres carry the metrics induced by a Riemannian metric on the total space that is Hermitian near the singular points of the central fibre. We prove that the $N-1$ eigenvalues of the Laplace–Beltrami operator on functions that tend to zero, multiplied by $\log(1/|s|)$, where s is the parameter, converge to 2π times the eigenvalues of the negative intersection matrix of the components relative to the diagonal matrix of their areas, and that the remaining positive eigenvalues stay bounded away from zero. Equivalently, the small eigenvalues are asymptotic to those of a finite weighted graph on the components whose edge weights are intersection numbers, in analogy with Burger's theorem for hyperbolic surfaces. This answers questions of Dai and Yoshikawa on the existence and geometric meaning of these limits, and evaluates the constant in their asymptotic formula for the product of the small eigenvalues as a weighted count of spanning trees. For Kähler metrics the limits can also be derived from recent work of Cao on the Archimedean height pairing combined with the eigenvalue bounds of Dai and Yoshikawa; the proof given here is direct. It uses the logarithms of local equations of the components as explicit harmonic functions on the smooth fibres, whose Dirichlet energies are computed by winding numbers.

1. Introduction

1.1. **The result.** Let X be a connected complex surface, $\mathbb{D} \subset \mathbb{C}$ a disc centred at the origin, and

$$\pi : X \longrightarrow \mathbb{D}$$

a proper surjective holomorphic map with connected fibres $X_s = \pi^{-1}(s)$, which is a submersion over $\mathbb{D} \setminus \{0\}$. We assume that the central fibre X_0 is a reduced divisor; equivalently, π has finitely many critical points, and these are the singular points of X_0: a reduced local equation of a curve has vanishing differential exactly at its singular points, and a multiple component would be critical along a curve. The fibre X_0 is connected, as a limit of connected fibres of a proper map. Let $C_1, \ldots, C_N$ be the irreducible components of X_0. No further condition is imposed on the singularities of X_0.

Let g be a smooth Riemannian metric on X that is Hermitian on a neighbourhood of $\operatorname{Sing} X_0$, and let g_s be its restriction to X_s for $s \neq 0$, and g_0 its restriction to $X_0 \setminus \operatorname{Sing} X_0$. For $s \neq 0$ let

$$0 = \lambda_0(s) < \lambda_1(s) \leq \lambda_2(s) \leq \cdots$$

be the eigenvalues of the Laplacian $\Delta_s = d^*d$ of (X_s, g_s) acting on functions. Let A_i be the area of $C_i \setminus \operatorname{Sing} X_0$ with respect to g.

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On $\mathbb{R}^N$ consider the two quadratic forms

$$q(F) = \sum_{i<j} (C_i \cdot C_j)\,(F_i - F_j)^2, \qquad m(F) = \sum_i A_i F_i^2, \tag{1}$$

where $C_i \cdot C_j$ is the intersection number in X. Since $\sum_i C_i = X_0$ is a fibre, $C_i \cdot X_0 = 0$, and therefore

$$q(F) = -\Big(\sum_i F_i C_i\Big)^2. \tag{2}$$

Thus q is the negative of the intersection form on the span of the components. It is positive semidefinite, and its kernel consists of the constant vectors because X_0 is connected; this is Zariski's lemma [BHPV04, Chapter III, Lemma 8.2], and in the present form it is also evident from (1). Let

$$0 = \mu_0 < \mu_1 \le \cdots \le \mu_{N-1}$$

be the eigenvalues of q relative to m, that is, the roots of $\det(Q - \mu\,\mathrm{diag}(A_i)) = 0$, where Q is the matrix of q.

Theorem 1.1. *With the preceding notation,*

$$\lim_{s\to 0} \log(1/|s|)\,\lambda_k(s) = 2\pi\mu_k \qquad (1 \le k \le N-1), \tag{3}$$

and $\liminf_{s\to 0} \lambda_N(s) > 0$.

The limit depends on the metric only through the areas of the components, and on the singularities of X_0 only through the local intersection multiplicities of distinct components. For a finite graph with weights (conductances) $c_e > 0$ on the edges and masses $m_v > 0$ at the vertices, we call the eigenvalues of the form $\sum_e c_e (F_{e^+} - F_{e^-})^2$ relative to $\sum_v m_v F_v^2$ the eigenvalues of the weighted graph. If X_0 has only ordinary nodes, the matrix of q is the Laplacian of the dual graph of X_0, and (3) says that the small eigenvalues are asymptotic to those of the dual graph with conductance $2\pi/\log(1/|s|)$ on each edge and mass A_i at each vertex. A point where two components are simply tangent counts as two edges, and an ordinary d-fold point through which d components pass contributes a complete graph on those components (Section 5).

Corollary 1.2. *If $N = 2$, then*

$$\lambda_1(s) = \frac{2\pi\,(C_1 \cdot C_2)\,(A_1 + A_2)}{A_1 A_2}\,\frac{1 + o(1)}{\log(1/|s|)}.$$

Corollary 1.3. *Let κ be the common value of the principal cofactors of the matrix of q; by the matrix-tree theorem* [Cha82] *it is the sum over the spanning trees $\mathcal{T}$ of the complete graph on $\{1,\dots,N\}$ of $\prod_{\{i,j\}\in\mathcal{T}} (C_i \cdot C_j)$. Then*

$$\prod_{k=1}^{N-1} \lambda_k(s) = \Big(\frac{2\pi}{\log(1/|s|)}\Big)^{N-1} \Big(\frac{\kappa \sum_i A_i}{\prod_i A_i} + o(1)\Big).$$

1.2. **Relation to earlier work.** For compact hyperbolic surfaces, Schoen, Wolpert and Yau [SWY80] bounded the small eigenvalues above and below by sums of lengths of short geodesics, and Burger [Bur90, Theorem 1.1] proved that they are asymptotic to the eigenvalues of a weighted graph: the vertices are the components of the complement of the short geodesics, with masses $2\gamma_v - 2 + n_v$, where γ_v is the genus and n_v the number of boundary geodesics and cusps of the component, and the edges are the geodesics, with weights their

lengths; the ratio of each small eigenvalue of the surface to the corresponding eigenvalue of the graph tends to $1/(2\pi^2)$.

For metrics induced from the total space, Dai and Yoshikawa [DY26] work in the setting above, with X a complex surface carrying a positive line bundle, g Kähler and X_0 reduced and reducible. They prove that $\lambda_1, \dots, \lambda_{N-1}$ are bounded above and below by constant multiples of $1/\log(1/|s|)$ [DY26, Theorem 0.2], and that $(\log(1/|s|))^{N-1} \prod_{k=1}^{N-1} \lambda_k(s)$ has a positive limit [DY26, Theorem 0.3]; in particular $\lambda_1(s)\log(1/|s|)$ converges when $N = 2$. The proof compares ratios of analytic torsions with determinants of period integrals. The constant is not evaluated. They ask whether $\lambda_k(s)\log(1/|s|)$ converges when $N > 2$, for a geometric expression of the limit, and whether there is a finite weighted graph whose eigenvalues are asymptotic to the small eigenvalues [DY26, Problems 9.2 and 9.4]. Theorem 1.1 answers these questions in the setting of that paper: the graph has vertices the components, masses A_i and conductances $2\pi(C_i \cdot C_j)/\log(1/|s|)$, and Corollary 1.3 evaluates the constant in their product formula. Their Laplacian is $\bar\partial^*\bar\partial = \frac{1}{2}d^*d$, so in their normalization 2π is replaced by π.

Cao [Cao26b] extends the two-sided bounds to the scalar Laplacian on degenerations of Kähler manifolds of any dimension, and conjectures that the rescaled small eigenvalues converge to those of a weighted dual graph, without specifying the edge weights [Cao26b, Theorem 0.4 and Section 3.C]. Theorem 1.1 confirms this for degenerations of curves with reduced central fibre and identifies the edge weights as the intersection numbers $C_i \cdot C_j$. In [Cao26a] he studies, for degenerating curves, the Archimedean height pairing of two-forms, defined by solving $dd^c\varphi = -\alpha$ on the fibres (with d^c normalized as in that paper; see Remark 5.6), and proves that its logarithmic coefficient is given by the Moore–Penrose inverse of the intersection matrix of the components [Cao26a, Theorem 0.2]. That result concerns the Green operator on the small eigenspaces, and Theorem 1.1 the eigenvalues themselves. For Kähler metrics, which include the setting of [DY26], the limits (3) can also be derived from [Cao26a] and the bounds of [DY26] by a finite-dimensional argument that is carried out in neither paper; we sketch it in Remark 5.6. What is added here is a direct proof that does not use the Green operator or analytic torsion, the extension to metrics that are only Hermitian near $\operatorname{Sing} X_0$, and the explicit identification of the graph and of the constants.

Ji and Wentworth [JW92] prove spectral convergence for plumbing families with the Bergman metric and, for a separating node, suggest a heuristic for the rate of λ_1 [JW92, Remark 5.10]; see Remark 5.7.

1.3. **Method.** Near a singular point p of X_0 write $\pi = \prod_i f_i$, where f_i is a local equation of C_i. On a smooth fibre the functions $\log|f_i|$ are harmonic and sum to $\log|s|$. For $F \in \mathbb{R}^N$ the function

$$h_F = \frac{1}{\log|s|} \sum_i F_i \log|f_i|$$

is harmonic, takes values between $\min F_i$ and $\max F_i$, and is close to F_i on the part of the fibre near C_i and away from p. By Stokes' theorem its Dirichlet energy is a sum of boundary terms, each the product of a boundary value F_i and a winding number of some f_j along a boundary circle. These winding numbers are local intersection multiplicities, and the energy is $2\pi/\log(1/|s|)$ times the local contribution to $q(F)$, up to an error of order $|F|^2/\log(1/|s|)^2$ (Lemma 2.4). Used as test functions the h_F give the upper bound; they play the role of the capacitary potentials of the neighbourhoods of the singular points in the fibre. For the lower bound we pair the differential of an arbitrary function of small energy with $d^c h_F$, where $d^c = i(\bar\partial - \partial)$; the Cauchy–Schwarz inequality then bounds its energy below by the same

quantity (Lemma 4.3). No resolution of the singularities of X_0 and no semistable reduction is used.

1.4. **Conventions.** We write $d^c = i(\bar\partial - \partial)$. On a Riemann surface with a conformal metric, $d^c u = *du$ for real functions u, so that $du \wedge d^c u = |du|^2\, dA$, and a function u is harmonic if and only if $dd^c u = 0$. If f is holomorphic and nowhere zero, $d^c \log|f| = d\arg f$. The Dirichlet energy of u over a domain M is $E_M(u) = \int_M |du|^2\, dA$, with associated bilinear form $E_M(u,v)$; we write $E = E_{X_s}$. Norms without subscript are L^2 norms on (X_s, g_s). The letter C denotes constants that depend only on (X, π, g), and $c(r)$ denotes constants that depend in addition on a radius r; neither depends on s, and their values may change from line to line. For $F \in \mathbb{R}^N$ we put $|F| = \max_i |F_i|$.

2. Local structure near the singular points

2.1. **Coordinates and local equations.** Let $\Sigma = \operatorname{Sing} X_0$. For each $p \in \Sigma$ choose holomorphic coordinates $z = (z_1, z_2)$ centred at p on an open set U_p on which g is Hermitian, such that neither $\pi(z_1, 0)$ nor $\pi(0, z_2)$ vanishes identically. Write $B_p(r) = \{|z| < r\}$ and $S_p(r) = \partial B_p(r)$. Since X_0 is reduced, the germ of π at p is a product of pairwise non-associated irreducible germs, one for each local branch β of X_0 at p. Collecting the branches that lie on the same component, and absorbing the unit into one factor, we obtain holomorphic functions $f_1, \dots, f_N$ on U_p, after shrinking it, with

$$\pi = \prod_{i=1}^{N} f_i \quad \text{on } U_p, \qquad \{f_i = 0\} = C_i \cap U_p, \tag{4}$$

where f_i vanishes to first order along $C_i \cap U_p$ and $f_i \equiv 1$ if $p \notin C_i$. Shrinking U_p again, $|f_i| < 1$ whenever $p \in C_i$. For a branch β let $i(\beta)$ be the index of the component containing it, and for $j \neq i(\beta)$ let $(\beta \cdot C_j)_p$ be the intersection multiplicity of β with C_j at p, that is, the order of vanishing at p of f_j along the normalization of β. Then

$$\sum_{\beta \subset C_i} (\beta \cdot C_j)_p = (C_i \cdot C_j)_p, \qquad \sum_{p \in \Sigma} (C_i \cdot C_j)_p = C_i \cdot C_j \qquad (i \neq j). \tag{5}$$

2.2. **Radii.** Each branch β of X_0 at p has an injective holomorphic parametrization $\tau \mapsto \phi(\tau)$ on a disc, with $\phi(0) = p$ and $|\phi(\tau)|^2 = c|\tau|^{2n}(1 + O(|\tau|))$ for some $c > 0$ and $n \geq 1$, so that $|\phi|$ is strictly increasing along rays for small $|\tau|$. Hence there is $\rho_0 > 0$ such that for every $p \in \Sigma$ the ball $\overline{B_p(4\rho_0)}$ lies in U_p, these closed balls are pairwise disjoint, and for $0 < r \leq 4\rho_0$ the sphere $S_p(r)$ meets X_0 transversally, each branch β meeting $B_p(r)$ in the image of a disc $\{|\phi| < r\}$ that is star-shaped about 0, whose boundary circle $\gamma^0_\beta(r) = \beta \cap S_p(r)$ therefore winds once around p in the normalization of β; compare the conic structure theorem [Mil68, Section 2]. For $0 < r \leq 2\rho_0$ put

$$K_r = X \setminus \bigcup_{p \in \Sigma} B_p(r), \qquad \Omega^0_i(r) = C_i \cap K_r.$$

Each $\Omega^0_i(r)$ is a compact connected surface with boundary: its preimage in the normalization of C_i is the complement of finitely many discs. The sets $\Omega^0_i(r)$, $1 \leq i \leq N$, are pairwise disjoint.

Lemma 2.1. *For every $r \in (0, 2\rho_0]$ there are $\delta(r) > 0$ and a smooth map*

$$\Phi : (X_0 \cap K_r) \times \{|s| < \delta(r)\} \longrightarrow K_r,$$

a diffeomorphism onto $\pi^{-1}(\{|s|<\delta(r)\})\cap K_r$*, with* $\pi\circ\Phi(x,s)=s$ *and* $\Phi(x,0)=x$*. For* $0<|s|<\delta(r)$*:*

(1) $X_s\cap K_r$ *is the disjoint union of the connected surfaces* $\Omega_i(r,s)=\Phi(\Omega_i^0(r),s)$*, and the metrics* $\Phi(\cdot,s)^*g_s$ *converge to* g_0 *in every* C^k *norm on* $X_0\cap K_r$ *as* $s\to 0$*;*

(2) X_s *meets each* $S_p(r)$ *transversally, in a disjoint union of circles* $\gamma_\beta(r,s)$*, one for each branch* β *at* p*, with* $\gamma_\beta(r,s)\to\gamma_\beta^0(r)$ *in* C^1 *as* $s\to 0$*;*

(3) $M_p(r,s)=X_s\cap\overline{B_p(r)}$ *is a compact surface with boundary* $\bigcup_\beta\gamma_\beta(r,s)$*.*

Proof. The map π has no critical points on $X_0\cap K_r$, and its restriction to the boundary spheres has none on $X_0\cap\partial K_r$, by transversality. The set of points of K_r at which π or $\pi|_{\partial K_r}$ fails to be a submersion is closed and disjoint from X_0; since π is proper, its image misses a disc $\{|s|<\delta(r)\}$. Over this disc the real coordinate vector fields of $\mathbb{D}$ have lifts to K_r that are tangent to ∂K_r: such lifts exist locally, at boundary points because $\pi|_{\partial K_r}$ is a submersion, and the two conditions are preserved by a partition of unity. Composing the flows of the two lifts gives Φ, as in Ehresmann's theorem. Assertion (1) follows because Φ is smooth and $\Phi(\cdot,0)$ is the identity. Since $\pi|_{S_p(r)}$ is a submersion along $X_s\cap S_p(r)$, the fibre X_s is transverse to $S_p(r)$, and $X_s\cap S_p(r)=\Phi(X_0\cap S_p(r),s)$; this gives (2) and (3). □

We always take $|s|$ smaller than δ of every radius involved, without further mention. The maps Φ for different radii need not agree, but the sets they define are compatible for small s: $\Omega_i(r,s)$ is the unique connected component of $X_s\cap K_r$ contained in a prescribed small neighbourhood of $\Omega_i^0(r)$ in X, because Φ^{-1} is continuous and the $\Omega_j^0(r)$ are compact and pairwise disjoint. Consequently, for $r'<r$ and small s, $\Omega_i(r,s)\subset\Omega_i(r',s)$, the circle $\gamma_\beta(r,s)$ lies in the interior of $\Omega_{i(\beta)}(r',s)$ and is a boundary component of $\Omega_{i(\beta)}(r,s)$, and both lie in the set W_β of Lemma 2.3 below.

Lemma 2.2. *After decreasing* ρ_0 *there is* C *such that* $\operatorname{Area}(X_s\cap B_p(r))\le Cr^2$ *for all* $p\in\Sigma$*, all* $r\in(0,4\rho_0]$ *and all sufficiently small* s*, including* $s=0$*. Consequently*

$$A_i - Cr^2 \le \lim_{s\to 0}\operatorname{Area}\Omega_i(r,s) = \operatorname{Area}\Omega_i^0(r)\le A_i. \tag{6}$$

Proof. Let n_2 be the order of vanishing of $\pi(0,z_2)$ at $z_2=0$. Choose $\rho''>0$ with $\pi(0,z_2)\neq 0$ for $0<|z_2|\le\rho''$. By Rouché's theorem there are $\rho'>0$ and $\delta'>0$ such that for $|z_1|\le\rho'$ and $|s|\le\delta'$ the function $z_2\mapsto\pi(z_1,z_2)-s$ has exactly n_2 zeros in $\{|z_2|<\rho''\}$, counted with multiplicity. Thus, if $4\rho_0\le\min(\rho',\rho'')$, the projection of $X_s\cap B_p(r)$ to the z_1-axis has image in the disc of radius r and at most n_2 points in each fibre, so $\int_{X_s\cap B_p(r)}\frac{i}{2}dz_1\wedge d\bar z_1\le n_2\pi r^2$. The same holds with the two coordinates exchanged. The sum of the two integrals is the area of the complex curve $X_s\cap B_p(r)$ for the flat metric of the chart, and g is bounded by a constant multiple of that metric on $B_p(4\rho_0)$. The convergence in (6) is Lemma 2.1(1), and the inequalities follow from the first assertion at $s=0$. □

2.3. **The harmonic functions.** Fix $p\in\Sigma$. For $s\neq 0$ the functions f_i of (4) are holomorphic and nowhere zero on $X_s\cap U_p$, and $\sum_i\log|f_i|=\log|s|$ there. For $F\in\mathbb{R}^N$ define

$$h_F=\frac{1}{\log|s|}\sum_{i=1}^N F_i\log|f_i| \qquad \text{on } X_s\cap U_p. \tag{7}$$

It is harmonic and linear in F. The numbers $t_i=\log|f_i|/\log|s|$ are nonnegative for $|s|<1$ and sum to one, so

$$\min_i F_i\le h_F\le\max_i F_i. \tag{8}$$

Lemma 2.3. *Let $0 < r \le \rho_0$. For every branch β at p, with $i = i(\beta)$, there is a neighbourhood W_β in X of $\beta \cap (\overline{B_p(4\rho_0)} \setminus B_p(r/2))$ such that, for all sufficiently small $s \neq 0$, the set $X_s \cap (\overline{B_p(4\rho_0)} \setminus B_p(r/2))$ is contained in the disjoint union of the W_β, and on $X_s \cap W_\beta$*

$$h_F - F_i = \frac{1}{\log|s|} \sum_{j \neq i} (F_j - F_i) \log|f_j|, \qquad |h_F - F_i| + |dh_F|_{g_s} \le \frac{c(r)\,|F|}{\log(1/|s|)}. \tag{9}$$

Proof. For $j \neq i$ the function f_j vanishes only on C_j, which meets β in $B_p(4\rho_0)$ only at p. Hence $\log|f_j|$ is smooth and bounded with bounded differential on a neighbourhood W_β of the compact set $\beta \cap (\overline{B_p(4\rho_0)} \setminus B_p(r/2))$, and these neighbourhoods may be taken pairwise disjoint. By Lemma 2.1 the fibre X_s lies in their union over the given region for small s. The identity in (9) follows from $\log|f_i| = \log|s| - \sum_{j \neq i} \log|f_j|$, and the estimate from the identity. □

Define the local forms

$$q_p(F,G) = \sum_{i<j} (C_i \cdot C_j)_p\, (F_i - F_j)(G_i - G_j), \qquad q = \sum_{p \in \Sigma} q_p, \tag{10}$$

the second identity being (5); here $q(F,G)$ is the bilinear form of (1).

Lemma 2.4 (Energy identity)**.** *Let $0 < r \le \rho_0$ and $M = M_p(r,s)$. For $F, G \in \mathbb{R}^N$,*

$$\sum_\beta F_{i(\beta)} \int_{\gamma_\beta(r,s)} d^c h_G = \frac{2\pi}{\log(1/|s|)}\, q_p(F,G), \tag{11}$$

where the circles carry the orientation induced from M, and

$$\left| E_M(h_F, h_G) - \frac{2\pi}{\log(1/|s|)}\, q_p(F,G) \right| \le \frac{c(r)\,|F|\,|G|}{\log(1/|s|)^2}. \tag{12}$$

Proof. Let β be a branch with $i = i(\beta)$, and let $j \neq i$. On W_β the form $d^c \log|f_j| = d \arg f_j$ is closed, and by Lemma 2.1(2) the circle $\gamma_\beta(r,s)$ is homotopic in W_β to $\gamma_\beta^0(r)$, with matching orientations. The latter is the boundary of the disc $\beta \cap B_p(r)$ and winds once around p in the normalization of β, so the argument principle gives

$$\int_{\gamma_\beta(r,s)} d^c \log|f_j| = 2\pi\, (\beta \cdot C_j)_p \qquad (j \neq i).$$

Since $\log|f_i| = \log|s| - \sum_{j \neq i} \log|f_j|$ on X_s,

$$\int_{\gamma_\beta(r,s)} d^c h_G = \frac{2\pi}{\log|s|} \sum_{j \neq i} (G_j - G_i)\, (\beta \cdot C_j)_p.$$

Multiply by F_i and sum over the branches. By (5) the result is

$$\frac{2\pi}{\log|s|} \sum_i \sum_{j \neq i} F_i (G_j - G_i)(C_i \cdot C_j)_p = -\frac{2\pi}{\log|s|} \sum_{i<j} (C_i \cdot C_j)_p (F_i - F_j)(G_i - G_j),$$

which is (11). Since h_G is harmonic, Stokes' theorem gives

$$E_M(h_F, h_G) = \int_M dh_F \wedge d^c h_G = \sum_\beta \int_{\gamma_\beta(r,s)} h_F\, d^c h_G.$$

On $\gamma_\beta(r,s)$ we have $|h_F - F_{i(\beta)}| \le c(r)|F|/\log(1/|s|)$ and $|d^c h_G| \le c(r)|G|/\log(1/|s|)$ by Lemma 2.3, and the lengths of these circles are bounded. Replacing h_F by $F_{i(\beta)}$ and applying (11) proves (12). □

3. The upper bound

Proposition 3.1. *For $1 \le k \le N-1$, $\limsup_{s\to 0} \log(1/|s|)\,\lambda_k(s) \le 2\pi\mu_k$.*

Proof. Fix $r \in (0, \rho_0]$ and write $L = \log(1/|s|)$. For each $p \in \Sigma$ let χ_p be a smooth function on X with values in $[0,1]$, equal to one on $B_p(r)$ and supported in $B_p(2r)$. For $F \in \mathbb{R}^N$ define a function U_F on X_s by

$$U_F = \begin{cases} F_i & \text{on } \Omega_i(2r,s), \\ F_{i(\beta)} + \chi_p\,(h_F - F_{i(\beta)}) & \text{on } X_s \cap W_\beta \cap \overline{B_p(2r)}, \\ h_F & \text{on } M_p(r,s), \end{cases}$$

where h_F is the function (7) at p. By Lemmas 2.1 and 2.3 these prescriptions agree on overlaps for small s and define a smooth function, linear in F. The energy of U_F vanishes on the $\Omega_i(2r,s)$ and equals $E_{M_p(r,s)}(h_F)$ on $M_p(r,s)$. On the collars $X_s \cap (B_p(2r) \setminus B_p(r))$, which have bounded area, Lemma 2.3 gives $|dU_F| \le |d\chi_p||h_F - F_{i(\beta)}| + |dh_F| \le c(r)|F|/L$. Hence by Lemma 2.4

$$\left| E(U_F) - \frac{2\pi}{L}\, q(F) \right| \le \frac{c(r)|F|^2}{L^2}. \tag{13}$$

By (8), U_F is everywhere a convex combination of the F_i, so $|U_F| \le |F|$. Lemma 2.2 therefore gives

$$\left| \|U_F\|^2 - \sum_i F_i^2 \operatorname{Area} \Omega_i(2r,s) \right| \le Cr^2|F|^2, \qquad \liminf_{s\to 0} \|U_F\|^2 \ge m(F) - Cr^2|F|^2. \tag{14}$$

Let $V_k \subset \mathbb{R}^N$ be the span of eigenvectors of q relative to m for $\mu_0, \dots, \mu_k$. Note that $|F|^2 \le m(F)/\min_i A_i$, and that the convergence of the areas in (14) is uniform on the unit sphere of $\mathbb{R}^N$. If r is small, (14) shows that $F \mapsto U_F$ is injective on V_k for small s, so $\{U_F : F \in V_k\}$ has dimension $k+1$. On V_k we have $q(F) \le \mu_k m(F)$. The min–max principle, (13) and (14) give

$$\limsup_{s\to 0} L\,\lambda_k(s) \le \sup_{F \in V_k \setminus \{0\}} \frac{2\pi\, q(F)}{m(F) - Cr^2|F|^2} \le \frac{2\pi\mu_k}{1 - Cr^2/\min_i A_i}.$$

Let $r \to 0$. □

4. The lower bound

Throughout this section u denotes an eigenfunction of Δ_s with eigenvalue $\lambda \le 1$ and $\|u\| = 1$, and $r \in (0, \rho_0]$ is fixed. Let $a_i = a_i(u)$ be the mean value of u over $\Omega_i(r/2, s)$.

Lemma 4.1. *For sufficiently small s,*

1. $|a_i| \le C$;
2. $\|u - a_i\|_{L^2(\Omega_i(r/2,s))} \le c(r)\lambda^{1/2}$;
3. $\sup_{\gamma_\beta(r,s)} |u - a_{i(\beta)}| \le c(r)\lambda^{1/2}$ *for every branch β at every $p \in \Sigma$.*

Proof. By the Cauchy–Schwarz inequality, $|a_i| \le \operatorname{Area}(\Omega_i(r/2,s))^{-1/2}$, and (1) follows from (6) once ρ_0 is small. For (2) and (3) transport u to the fixed surface $\Omega_i^0(r/2)$ by the diffeomorphisms of Lemma 2.1. The transported metrics converge smoothly to g_0, so the Poincaré inequality on the connected surface $\Omega_i^0(r/2)$ holds for them with a uniform constant; since $E(u) = \lambda$, this is (2). The function $v = u - a_i$ satisfies $\Delta_s v = \lambda u$, whose L^2 norm is at most λ.

The curves $\gamma_\beta(r,s)$ with $i(\beta)=i$ remain in a fixed compact subset of the interior of $\Omega_i^0(r/2)$ after transport. Interior elliptic estimates with uniformly controlled coefficients and the Sobolev embedding $W^{2,2}\subset C^0$ in dimension two give $\sup_{\gamma_\beta}|v|\le c(r)(\|\Delta_s v\|_{L^2}+\|v\|_{L^2})\le c(r)\lambda^{1/2}$, the norms being taken over $\Omega_i(r/2,s)$. □

Lemma 4.2. *For sufficiently small s and every $p\in\Sigma$,*

$$\int_{M_p(r,s)} u^2\,dA \le Cr^2 + c(r)\lambda^{1/2}.$$

Proof. Let $M=M_p(r,s)$ and let h be the harmonic function on M with boundary values u. Then $w=u-h$ vanishes on ∂M, and $E_M(w)\le E_M(u)\le\lambda$, because $E_M(u)=E_M(h)+E_M(w)$ for h harmonic and w vanishing on the boundary. By the maximum principle and Lemma 4.1, $|h|\le C+c(r)\lambda^{1/2}$, so Lemma 2.2 gives $\int_M h^2\,dA\le Cr^2(C+c(r)\lambda^{1/2})^2$.

For w we compare with the flat metric of the chart. Let dA_0 be the area form on $X_s\cap U_p$ induced by that metric. Both induced metrics are Hermitian on the Riemann surface $X_s\cap U_p$, hence conformal; the Dirichlet energy is the same for both, and $dA\le C\,dA_0$. Since $dd^c|z|^2$ restricts to $4\,dA_0$ on a complex curve, Stokes' theorem gives

$$4\int_M w^2\,dA_0 = -2\int_M w\,dw\wedge d^c|z|^2 \le 4r\Big(\int_M w^2\,dA_0\Big)^{1/2} E_M(w)^{1/2},$$

because $|d|z|^2|\le 2|z|\le 2r$ on M. Hence $\int_M w^2\,dA\le Cr^2\lambda$. The lemma follows from $u^2\le 2h^2+2w^2$. □

Lemmas 4.1 and 4.2 extend to linear combinations $u=\sum_{l=1}^n\zeta_l u_l$ of orthonormal eigenfunctions with eigenvalues at most $\lambda\le 1$ and $\sum_l\zeta_l^2=1$, with constants multiplied by a factor depending only on n: the means $a_i(u)$ are linear in u, the estimates of Lemma 4.1 transfer by the triangle inequality, and $E(u)=\sum_l\zeta_l^2\lambda_l\le\lambda$, which is all that the proof of Lemma 4.2 uses besides them.

Lemma 4.3 (Energy lower bound)**.** *For sufficiently small s, every $p\in\Sigma$ and $a=(a_i(u))_i$,*

$$E_{M_p(r,s)}(u)\ge \frac{2\pi}{\log(1/|s|)}\,q_p(a) - \frac{c(r)}{\log(1/|s|)}\Big(\lambda^{1/2}+\frac{1}{\log(1/|s|)}\Big).$$

The same inequality holds, with $c(r)$ multiplied by a constant depending only on n, for $u=\sum_{l=1}^n\zeta_l u_l$, where $u_1,\dots,u_n$ are orthonormal eigenfunctions with eigenvalues at most λ, $\sum_l\zeta_l^2=1$, and $a=\sum_l\zeta_l a(u_l)$.

Proof. Write $M=M_p(r,s)$ and $L=\log(1/|s|)$, and let h_a be the function (7) for $F=a$. It is harmonic and smooth up to ∂M, so

$$I:=\int_M du\wedge d^c h_a = \sum_\beta a_{i(\beta)}\int_{\gamma_\beta(r,s)} d^c h_a + \sum_\beta\int_{\gamma_\beta(r,s)}(u-a_{i(\beta)})\,d^c h_a.$$

The first sum equals $P:=(2\pi/L)\,q_p(a)\ge 0$ by (11). By Lemmas 2.3 and 4.1, the second is bounded in absolute value by $e:=c(r)\lambda^{1/2}/L$. By Lemma 2.4, $E_M(h_a)\le P+\theta$ with $\theta=c(r)/L^2$. If $P\le e$ there is nothing to prove. Otherwise $I\ge P-e>0$, and the Cauchy–Schwarz inequality $I^2\le E_M(u)E_M(h_a)$ gives

$$E_M(u)\ge\frac{(P-e)^2}{P+\theta}\ge P-2e-\theta.$$

For a combination $u=\sum\zeta_l u_l$ the same argument applies, by the paragraph preceding the lemma. □

Proposition 4.4. *For $1 \le k \le N-1$,*

$$\liminf_{s\to 0} \log(1/|s|)\,\lambda_k(s) \ge 2\pi\mu_k.$$

Moreover $\liminf_{s\to 0}\lambda_N(s) > 0$.

Proof. Let $n \le N+1$ and let $u_1,\dots,u_n$ be orthonormal eigenfunctions of Δ_s with eigenvalues at most $\lambda \le 1$. Let V be their span and $\Theta : V \to \mathbb{R}^N$ the linear map $u \mapsto (a_i(u))_i$. For $u = \sum \zeta_l u_l$ with $\sum \zeta_l^2 = 1$, the triangle inequality in L^2 and Lemmas 4.1 and 4.2 give

$$\int_{\Omega_i(r,s)} u^2\,dA \le a_i^2 \operatorname{Area}\Omega_i(r,s) + c(r)\lambda^{1/2}, \qquad \int_{M_p(r,s)} u^2\,dA \le Cr^2 + c(r)\lambda^{1/2},$$

with constants depending on N. Since X_s is the union of the $\Omega_i(r,s)$ and the $M_p(r,s)$, summing and using (6) yields

$$m(\Theta u) \ge \big(1 - Cr^2 - c(r)\lambda^{1/2} - o(1)\big)\|u\|^2 \qquad (u \in V), \tag{15}$$

where $o(1) \to 0$ as $s \to 0$ for fixed r.

The gap. Suppose $\lambda_N(s_l) \to 0$ along a sequence $s_l \to 0$. Apply the preceding with $n = N+1$, the eigenfunctions for $\lambda_0,\dots,\lambda_N$ and $\lambda = \lambda_N(s_l)$. Choosing r with $Cr^2 < 1/2$, (15) shows that Θ is injective for large l, which is impossible since $\dim V = N+1$.

The limits. Let $1 \le k \le N-1$, $n = k+1$, and take the eigenfunctions for $\lambda_0,\dots,\lambda_k$, with $\lambda = \lambda_k(s)$. By Proposition 3.1, $\lambda \le C/L$ with $L = \log(1/|s|)$. For r small and then s small, (15) shows that Θ is injective, so $\Theta(V)$ has dimension $k+1$ and contains a vector Θu, $\|u\| = 1$, with $q(\Theta u) \ge \mu_k\, m(\Theta u)$. For this u, Lemma 4.3, summed over $p \in \Sigma$, and (15) give

$$\lambda_k(s) \ge E(u) \ge \sum_{p\in\Sigma} E_{M_p(r,s)}(u) \ge \frac{2\pi\mu_k}{L}\big(1 - Cr^2 - o(1)\big) - \frac{c(r)}{L^{3/2}}.$$

Multiply by L, let $s \to 0$ and then $r \to 0$. □

Propositions 3.1 and 4.4 prove Theorem 1.1.

Proof of Corollaries 1.2 and 1.3. Let Q be the matrix of q and $A = \operatorname{diag}(A_i)$. The μ_k are the roots of $\det(Q - tA)$. This polynomial has leading coefficient $(-1)^N \prod_i A_i$, no constant term, and linear coefficient $-\sum_i A_i \det Q^{(i)}$, where $Q^{(i)}$ is obtained by deleting the ith row and column. Since Q is a weighted graph Laplacian, all $\det Q^{(i)}$ are equal to the weighted number κ of spanning trees [Cha82]. Hence $\prod_{k\ge1}\mu_k = \kappa \sum_i A_i / \prod_i A_i$, and Corollary 1.3 follows from (3). For $N = 2$, $\kappa = C_1 \cdot C_2$. □

5. Examples and remarks

Example 5.1 (Nodal fibres)**.** If X_0 has only ordinary nodes, $C_i \cdot C_j$ is the number of nodes joining C_i and C_j, and nodes of a single component do not enter. The matrix of q is the Laplacian of the dual graph of X_0 with loops removed.

Example 5.2 (Tangency and higher contact)**.** Let two components be given near p by $f_1 = y$ and $f_2 = y - x^n$, with $\pi = f_1 f_2$. Then $(C_1 \cdot C_2)_p = n$: the point contributes as n nodes. We sketch a direct argument. On the fibre $\{y(y-x^n) = s\}$ the function y omits zero, and each of its values other than the two square roots of s is taken at the n solutions of $x^n = y - s/y$. After removing from the local fibre and from the annulus sets on which $\log|y|/\log|s|$ has energy $O(\log(1/|s|)^{-2})$, y therefore exhibits the local fibre as an n-sheeted branched cover of an annulus $\{c^{-1}|s| < |y| < c\}$. The function $\log|y|/\log|s|$ has energy $2\pi\log(c^2/|s|)/\log(1/|s|)^2$ on the annulus and n times as much on the cover.

Example 5.3 (Ordinary multiple points)**.** Suppose d distinct components pass through p with pairwise distinct tangents. Then q_p is the Laplacian form of the complete graph on these d components. This follows from $(C_i \cdot C_j)_p = 1$. It can be interpreted as follows, although the theorem does not apply to the non-reduced fibre of the blown-up family: in the blow-up of p the exceptional curve appears with multiplicity d and meets each strict transform once; the corresponding star-shaped network, with conductance d on each edge and no mass at the centre, has the complete graph with unit conductances as its star–mesh reduction to the outer vertices, that is, the Schur complement of its Laplacian with respect to the centre.

Example 5.4 (The Fermat pencil)**.** Let $d \geq 2$ and

$$X = \{(s, [x : y : z]) \in \mathbb{D} \times \mathbb{P}^2 : \ x^d + y^d + sz^d = 0\},$$

with π the first projection and g the restriction of the product of the flat metric on $\mathbb{D}$ and the Fubini–Study metric, so that g_s is the Fubini–Study metric on the plane curve X_s. This is [DY26, Example 8.1]. The surface X is smooth, and X_0 consists of d lines through $[0 : 0 : 1]$, with $\pi = -\prod_\zeta (x - \zeta y)$ in the coordinates (x, y) at that point, the product over the dth roots of -1. Thus $N = d$, $C_i \cdot C_j = 1$ for $i \neq j$, and all A_i equal the area A of a line. The Laplacian of the complete graph has the single nonzero eigenvalue d, of multiplicity $d - 1$, so

$$\lambda_k(s) = \frac{2\pi d}{A} \frac{1 + o(1)}{\log(1/|s|)} \qquad (1 \leq k \leq d - 1).$$

Remark 5.5 (Burger's theorem). The hyperbolic metrics on the fibres of a degenerating family of stable curves are not induced by a smooth metric on the total space, and Theorem 1.1 does not apply to them. Nevertheless, after the following conversion Burger's graph coincides with the graph of Theorem 1.1. At a node with plumbing parameter t, that is, with local equation $xy = t$ in the family, the short geodesic has length $\ell = 2\pi^2/\log(1/|t|) + O(\log(1/|t|)^{-2})$ [Wol90, Example 4.3], and the hyperbolic area of a component is $2\pi(2\gamma_v - 2 + n_v)$. Burger's ratio $1/(2\pi^2)$ for edge weight ℓ and vertex mass $2\gamma_v - 2 + n_v$ [Bur90, Theorem 1.1] therefore corresponds to conductance $2\pi/\log(1/|t|)$ per node and mass equal to the area, as in Example 5.1. Burger's theorem moreover allows the lengths to tend to zero at independent rates, which is outside the one-parameter setting considered here.

Remark 5.6 (The pairing of Cao). Let $\mathsf{M} = (C_i \cdot C_j)$ be the intersection matrix, so that $-\mathsf{M}$ is the matrix of q. For a Kähler metric on X and reduced X_0, Cao considers the pairing $\langle \alpha, \alpha' \rangle(s) = \int_{X_s} \varphi_s \alpha'$ of smooth two-forms with vanishing fibre integrals, where φ_s solves $dd^c \varphi_s = -\alpha|_{X_s}$ with his normalization $d^c = \frac{i}{4\pi}(\bar\partial - \partial)$, and proves that $\langle \alpha, \alpha' \rangle - c_{\alpha,\alpha'} \log |s|^2$ extends continuously to $\mathbb{D}$, with $c_{\alpha,\alpha'} = v_\alpha^T \mathsf{M}^+ v_{\alpha'}$, $v_\alpha = (\int_{C_i} \alpha)_i$, and M^+ the Moore–Penrose inverse [Cao26a, Theorem 0.2]. Since $\langle \alpha, \alpha \rangle$ equals $4\pi \sum_{k \geq 1} \lambda_k^{-1} (\int_{X_s} \Phi_k \alpha)^2$ for an orthonormal basis of eigenfunctions Φ_k, the logarithmic term is carried by the $N - 1$ small eigenvalues. Cao also shows that the corresponding eigenfunctions are, up to errors of order $\log(1/|s|)^{-1/2}$ in L^2, combinations of functions that are constant on the parts of X_s near the components [Cao26a, Theorem 2.10]. We sketch how the limits (3) follow. The period vectors v_α exhaust the vectors with zero sum, and from these statements and the two-sided bounds of [DY26] one can deduce, by comparing two quadratic forms on that space along subsequences, that $\lambda_k(s) \log(1/|s|)$ converges to $2\pi\mu_k$. Thus for Kähler metrics the limits (3) also follow from [Cao26a] together with [DY26], although they are not stated there; the two results describe the same leading behaviour of the Green operator on the small eigenspaces from complementary sides.

Remark 5.7 (A heuristic of Ji and Wentworth). For a plumbing family with a separating node and the Bergman metric, which is not induced from the total space, [JW92, Remark 5.10] suggests that λ_1 should behave like the first Dirichlet eigenvalue of one of the two components with a disc of radius $\varepsilon \sim |t|^{1/2}$ removed, namely $2\pi/(A_i \log(1/\varepsilon))$ to leading order, and concludes that λ_1 is expected to be bounded above by a multiple of $1/\log(1/|t|)$. Taken literally the heuristic gives $4\pi/(A_i \log(1/|t|))$. For induced metrics, Corollary 1.2 gives $2\pi(1/A_1 + 1/A_2)/\log(1/|t|)$ at a single node, which coincides with that value exactly when $A_1 = A_2$.

Remark 5.8 (Hypotheses). The Hermitian condition on g is used only in the balls $B_p(4\rho_0)$, where it makes the Dirichlet energy on the fibres conformally invariant and the functions h_F harmonic. If X_0 is not reduced the functions (7) are still defined, with f_i replaced by $f_i^{m_i}$ for the multiplicities m_i, but the fibre near a multiple component is a covering of it that need not be connected, and according to [Cao26b, Theorem 0.4] the number of small eigenvalues is the number of components of the central fibre after a base change and normalization, minus one; we do not treat this case. In higher dimension, for two-forms on Calabi–Yau threefolds in conifold transitions with Ricci-flat metrics, it is shown in the preprint [W26] that the small eigenvalues have the same rate and the same constant 2π per node, with the intersection form on divisor classes in the role of m and the periods over the exceptional curves in the role of the differences $F_i - F_j$.

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